# Exercise 2: Vector calculus, conservation laws, classification

**Prognostic Modelling (pea700), WiSe 2026/27**

*Out: week 2 · Back: week 3 · 15 points*

Covers unit 2 of the script. Hand in your solutions in the folder `ex02/` of your repository, as PDF or in the markdown cells of a notebook.

## Problem 1: Gradient (2 points)

The temperature on a plate is $T(x, y) = 20 - x^2 - 2y^2$ (in °C, $x$ and $y$ in m).

a) Compute the gradient at the point $(1, 1)$ and the unit vector in its direction.

b) In which direction does heat flow at this point according to Fourier's law, and what is the largest rate of change of temperature with distance?

### Solution

a) The partial derivatives are taken with the other variable held fixed. The constant 20 and the term in the other variable drop out:

$$\frac{\partial T}{\partial x} = 0 - 2x - 0 = -2x, \qquad \frac{\partial T}{\partial y} = 0 - 0 - 4y = -4y.$$

Hence

$$\nabla T = \frac{\partial T}{\partial x}\vec{i} + \frac{\partial T}{\partial y}\vec{j} = -2x\,\vec{i} - 4y\,\vec{j}, \qquad \nabla T(1,1) = -2\vec{i} - 4\vec{j} \quad (\text{°C/m}).$$

Its magnitude is

$$\|\nabla T(1,1)\| = \sqrt{(-2)^2 + (-4)^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5},$$

and the unit vector in its direction is the gradient divided by its magnitude:

$$\vec{e} = \frac{\nabla T}{\|\nabla T\|} = \frac{-2\vec{i} - 4\vec{j}}{2\sqrt{5}} = \frac{-\vec{i} - 2\vec{j}}{\sqrt{5}}.$$

Check: $\|\vec{e}\|^2 = (1+4)/5 = 1$. Both components are negative: the temperature increases towards the origin, where $T$ has its maximum of 20 °C.

b) Fourier's law gives the heat flux $\vec{F} = -k\nabla T$ with $k > 0$. Because of the minus sign the flux points opposite to the gradient, from warm to cold:

$$\vec{F}(1,1) = -k\left(-2\vec{i} - 4\vec{j}\right) = k\left(2\vec{i} + 4\vec{j}\right), \qquad \text{direction} \quad -\vec{e} = \frac{\vec{i} + 2\vec{j}}{\sqrt{5}}.$$

Heat flows away from the warm centre, outward and more strongly in $y$ than in $x$.

The rate of change of $T$ in the direction of a unit vector $\vec{n}$ is the directional derivative

$$\frac{\mathrm{d}T}{\mathrm{d}s} = \nabla T \cdot \vec{n} = \|\nabla T\|\,\|\vec{n}\| \cos\alpha = \|\nabla T\| \cos\alpha,$$

where $\alpha$ is the angle between $\nabla T$ and $\vec{n}$. It is largest for $\alpha = 0$, that is in the direction of the gradient, and its largest value is

$$\|\nabla T(1,1)\| = 2\sqrt{5} \approx 4.47 \text{ °C/m}.$$

In the direction of the heat flux, $\alpha = \pi$, the temperature decreases at the same rate.

Points: a) 1, b) 1.

## Problem 2: Laplacian (3 points)

Let $f(x,y) = axy + ax^2 + by^3 + y^3$ with constants $a$ and $b$. Compute $\nabla \cdot (\nabla f)$. For which $a$ and $b$ does $f$ satisfy the Laplace equation everywhere?

**Solution**

**Gradient.** Term by term, holding $y$ fixed for $\partial/\partial x$ and $x$ fixed for $\partial/\partial y$:

$$\frac{\partial f}{\partial x} = ay + 2ax + 0 + 0 = ay + 2ax,$$
$$\frac{\partial f}{\partial y} = ax + 0 + 3by^2 + 3y^2 = ax + 3(b+1)\,y^2,$$

so $\nabla f = (ay + 2ax)\,\vec{i} + \left(ax + 3(b+1)y^2\right)\vec{j}$.

**Divergence of the gradient.** The divergence of $\vec{F} = F_1\vec{i} + F_2\vec{j}$ is $\partial F_1/\partial x + \partial F_2/\partial y$. With $F_1 = \partial f/\partial x$ and $F_2 = \partial f/\partial y$:

$$\frac{\partial}{\partial x}(ay + 2ax) = 0 + 2a = 2a, \qquad \frac{\partial}{\partial y}(ax + 3(b+1)y^2) = 0 + 6(b+1)\,y,$$

and

$$\nabla \cdot (\nabla f) = \nabla^2 f = 2a + 6(b+1)\,y.$$

The mixed term $axy$ does not contribute: its second derivatives $\partial^2(xy)/\partial x^2$ and $\partial^2(xy)/\partial y^2$ are both zero.

**Conclusion.** The Laplace equation $\nabla^2 f = 0$ must hold at every point, that is for all $y$. A polynomial in $y$ is zero for all $y$ only if all its coefficients are zero:

$$y^0:\ 2a = 0\ \Rightarrow\ a = 0, \qquad y^1:\ 6(b+1) = 0\ \Rightarrow\ b = -1.$$

With these values $f = 0 + 0 - y^3 + y^3 = 0$: only the trivial function remains. (A choice such as $a = -3y(b+1)$ is not admissible, because $a$ is a constant.)

Points: gradient 1, Laplacian 1, conclusion 1.

## Problem 3: Divergence of a radial field (3 points)

Consider the vector field

$$\vec{E} = \frac{\vec{r}}{\|\vec{r}\|^p}, \qquad \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}.$$

a) Derive $\nabla \cdot \vec{E}$ in Cartesian coordinates.

b) For which $p$ is the field free of divergence away from the origin? Which physical fields have this form?

### Solution

a) We write $r = \|\vec{r}\| = (x^2 + y^2 + z^2)^{1/2}$. The components of $\vec{E}$ are

$$E_1 = x\, r^{-p}, \qquad E_2 = y\, r^{-p}, \qquad E_3 = z\, r^{-p}.$$

**Derivative of $r$.** By the chain rule,

$$\frac{\partial r}{\partial x} = \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \cdot 2x = \frac{x}{r},$$

and in the same way $\partial r/\partial y = y/r$ and $\partial r/\partial z = z/r$.

**Derivative of $r^{-p}$.** Again by the chain rule, with the result above,

$$\frac{\partial}{\partial x} r^{-p} = -p\, r^{-p-1}\, \frac{\partial r}{\partial x} = -p\, r^{-p-1}\, \frac{x}{r} = -p\, x\, r^{-p-2}.$$

**Derivative of $E_1$.** By the product rule for $x \cdot r^{-p}$,

$$\frac{\partial E_1}{\partial x} = \frac{\partial x}{\partial x}\, r^{-p} + x\, \frac{\partial r^{-p}}{\partial x} = r^{-p} + x\left(-p\, x\, r^{-p-2}\right) = \frac{1}{r^p} - \frac{p\, x^2}{r^{p+2}}.$$

Correspondingly,

$$\frac{\partial E_2}{\partial y} = \frac{1}{r^p} - \frac{p\, y^2}{r^{p+2}}, \qquad \frac{\partial E_3}{\partial z} = \frac{1}{r^p} - \frac{p\, z^2}{r^{p+2}}.$$

**Sum.** Adding the three and using $x^2 + y^2 + z^2 = r^2$,

$$\nabla \cdot \vec{E} = \frac{3}{r^p} - \frac{p\,(x^2+y^2+z^2)}{r^{p+2}} = \frac{3}{r^p} - \frac{p\,r^2}{r^{p+2}} = \frac{3}{r^p} - \frac{p}{r^p} = \frac{3-p}{r^p}.$$

Check: for $p = 0$ the field is $\vec{E} = \vec{r}$, and directly $\nabla \cdot \vec{r} = 1 + 1 + 1 = 3$, which agrees with $(3 - 0)/r^0 = 3$.

b) $\nabla \cdot \vec{E} = 0$ for $r \neq 0$ requires $3 - p = 0$, so $p = 3$. Then

$$\vec{E} = \frac{\vec{r}}{r^3} = \frac{1}{r^2}\,\frac{\vec{r}}{r}, \qquad \|\vec{E}\| = \frac{1}{r^2}\,:$$

a field that points radially outward and whose magnitude decreases as $1/r^2$. This is the field of a point source: the gravitational field of a point mass (with a minus sign, it points inward) and the electric field of a point charge.

Interpretation with the divergence theorem: on a sphere of radius $R$ the outward normal is $\vec{r}/r$, so $\vec{E} \cdot \vec{r}/r = r^2/r^{p+1} = R^{1-p}$ on the sphere. The flux through the sphere is this value times the area $4\pi R^2$:

$$\oint_S \vec{E} \cdot \mathrm{d}\vec{A} = R^{1-p} \cdot 4\pi R^2 = 4\pi R^{3-p}.$$

It is independent of $R$ exactly for $p = 3$: the flux through every sphere around the origin is the same, $4\pi$, so no flux is created or lost between two spheres, and the divergence there is zero. The whole flux originates in the point source at the origin, where the field is not defined.

Points: a) 2, b) 1.

## Problem 4: Deriving a conservation law (4 points)

A dissolved substance with concentration $c(x, t)$ (kg/m³) is transported in a river with constant cross-section $A$ and constant velocity $v$. It is mixed by turbulence, described as diffusion with coefficient $D$, and it decays at the rate $\lambda c$ (kg/m³/s).

a) Write down the mass balance for the river section between $x$ and $x + \Delta x$, and derive the differential equation for $c$ by taking the limit $\Delta x \to 0$.

b) Classify the equation: order, linear or nonlinear, type.

c) Which initial and boundary conditions are needed for a river section of length $L$? Propose physically meaningful ones.

**Solution**

a) **Convention.** $x$ points downstream, $v > 0$, and the flux $F$ (kg/m$^2$/s) is counted positive in the $+x$ direction.

**Flux law.** The flux is the sum of the advective flux $vc$ and the diffusive flux from Fick's law, $-D\,\partial c/\partial x$ (down the gradient):

$$F = vc - D\frac{\partial c}{\partial x}.$$

**Balance.** The mass in the section is $c\,A\,\Delta x$ (for small $\Delta x$, with $c$ taken at a point inside the section). It changes by

- the inflow through the cross-section at $x$: $+A\,F(x)$, because a positive $F$ carries mass into the section;
- the outflow through the cross-section at $x + \Delta x$: $-A\,F(x + \Delta x)$, because a positive $F$ carries mass out of the section;
- decay inside the section: $-\lambda c\,A\,\Delta x$.

$$\frac{\partial}{\partial t}(c\,A\,\Delta x) = A\,F(x) - A\,F(x + \Delta x) - \lambda c\,A\,\Delta x.$$

**Limit.** $A$ and $\Delta x$ do not depend on time, so they can be taken out of the time derivative. Dividing by $A\Delta x$,

$$\frac{\partial c}{\partial t} = -\frac{F(x + \Delta x) - F(x)}{\Delta x} - \lambda c.$$

Note the order in the numerator: we wrote $F(x) - F(x + \Delta x) = -[F(x + \Delta x) - F(x)]$ to obtain a forward difference quotient. For $\Delta x \to 0$ it becomes the derivative:

$$\frac{\partial c}{\partial t} = -\frac{\partial F}{\partial x} - \lambda c.$$

**Inserting the flux law.** With constant $v$ and $D$,

$$\frac{\partial F}{\partial x} = \frac{\partial}{\partial x}(vc) - \frac{\partial}{\partial x}\left(D\frac{\partial c}{\partial x}\right) = v\frac{\partial c}{\partial x} - D\frac{\partial^2 c}{\partial x^2},$$

and therefore, with the minus sign in front distributed over both terms,

$$\frac{\partial c}{\partial t} = -v\frac{\partial c}{\partial x} + D\frac{\partial^2 c}{\partial x^2} - \lambda c,$$

or, with the advection term on the left,

$$\frac{\partial c}{\partial t} + v\frac{\partial c}{\partial x} = D\frac{\partial^2 c}{\partial x^2} - \lambda c.$$

Sign checks: without transport ($v = D = 0$) the equation is $\mathrm{d}c/\mathrm{d}t = -\lambda c$ with the solution $c = c_0 e^{-\lambda t}$, which decays for $\lambda > 0$. The diffusion term $D\,\partial^2 c/\partial x^2$ has a positive sign, so it lowers $c$ at a maximum ($\partial^2 c/\partial x^2 < 0$) and raises it at a minimum, as diffusion must.

b) **Order:** the highest derivative is $\partial^2 c/\partial x^2$: second order (first order in time).

**Linear:** $c$ and its derivatives appear only to the first power, are not multiplied with each other, and the coefficients $v$, $D$, $\lambda$ do not depend on $c$. There is no term independent of $c$, so it is also homogeneous.

**Type:** we bring all terms to one side,

$$D\frac{\partial^2 c}{\partial x^2} - \frac{\partial c}{\partial t} - v\frac{\partial c}{\partial x} - \lambda c = 0,$$

and compare with the form $a\,u_{xx} + 2b\,u_{xt} + \tilde{c}\,u_{tt} + \ldots$ of the script, here with the variables $(x, t)$; we write $\tilde{c}$ for the third coefficient to avoid a clash with the concentration $c$. The coefficients of the second derivatives are $a = D$, $b = 0$ (no mixed derivative) and $\tilde{c} = 0$ (no second time derivative). Then

$$b^2 - a\tilde{c} = 0 - D \cdot 0 = 0 :$$

parabolic. The first-order terms do not enter the classification. If advection dominates over diffusion, that is if the Péclet number $vL/D$ is large, the equation behaves like the hyperbolic advection equation $\partial c/\partial t + v\,\partial c/\partial x = 0$.

c) **Initial condition:** one, because the equation is of first order in time: the concentration along the river at the start, $c(x, 0) = c_0(x)$ for $0 \le x \le L$.

**Boundary conditions:** two, one at each end, because it is of second order in space. A physically meaningful choice is

- at the inflow, $x = 0$: the concentration of the water entering the section, $c(0, t) = c_{\mathrm{in}}(t)$ (Dirichlet), for example from a measuring station upstream;
- at the outflow, $x = L$: $\partial c/\partial x(L, t) = 0$ (Neumann). It sets the diffusive flux $-D\,\partial c/\partial x$ to zero, so that the total flux there is the advective flux $vc$: the substance leaves with the flow, and no concentration is imposed on the water downstream, about which we know nothing.

Points: a) 2, b) 1, c) 1.

## Problem 5: Classification (3 points)

For each equation state whether it is an ODE or a PDE, its order, whether it is linear, and for the PDEs of second order the type.

1. $\dfrac{\partial u}{\partial t} = D\dfrac{\partial^2 u}{\partial x^2}$
2. $\dfrac{\partial^2 u}{\partial t^2} = c^2\dfrac{\partial^2 u}{\partial x^2}$
3. $\dfrac{\partial^2 \phi}{\partial x^2} + \dfrac{\partial^2 \phi}{\partial y^2} = -\rho(x, y)$
4. $\dfrac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + \dfrac{g}{L}\sin\theta = 0$
5. $\dfrac{\partial u}{\partial t} + u\dfrac{\partial u}{\partial x} = \nu\dfrac{\partial^2 u}{\partial x^2}$
6. $\dfrac{\partial \theta(\psi)}{\partial t} = \dfrac{\partial}{\partial z}\left[K(\psi)\left(\dfrac{\partial \psi}{\partial z} + 1\right)\right]$

### Solution

| | Equation | Kind | Order | Linear | Type |
|---|---|---|---|---|---|
| 1 | diffusion | PDE | 2 | yes | parabolic |
| 2 | wave | PDE | 2 | yes | hyperbolic |
| 3 | Poisson | PDE | 2 | yes (inhomogeneous) | elliptic |
| 4 | pendulum | ODE | 2 | no ($\sin\theta$) | – |
| 5 | Burgers | PDE | 2 | no ($u\,u_x$) | parabolic |
| 6 | Richards | PDE | 2 | no ($K$ and $\theta$ depend on $\psi$) | parabolic |

**Rule.** For the type we write each second-order PDE in the form $a\,u_{xx} + 2b\,u_{xy} + c\,u_{yy} +$ lower-order terms $= f$, with all terms on one side, and evaluate $b^2 - ac$: negative means elliptic, zero parabolic, positive hyperbolic. To avoid a clash with the wave speed $c$ in equation 2, we call the three coefficients $a$, $b$ and $\tilde{c}$ below.

1. Diffusion, variables $(x, t)$. Written as $D\,u_{xx} - u_t = 0$: $a = D$, $b = 0$, $\tilde{c} = 0$ (no $u_{tt}$). $b^2 - a\tilde{c} = 0$: **parabolic**. $u$ and its derivatives appear linearly: linear, and homogeneous.

2. Wave, variables $(x, t)$. Written as $c^2 u_{xx} - u_{tt} = 0$: $a = c^2$, $b = 0$, $\tilde{c} = -1$. Note the minus sign that moving $u_{tt}$ to the other side produces. Then

$$b^2 - a\tilde{c} = 0 - c^2 \cdot (-1) = c^2 > 0 :$$

**hyperbolic**. Linear and homogeneous.

3. Poisson, variables $(x, y)$. $\phi_{xx} + \phi_{yy} = -\rho$: $a = 1$, $b = 0$, $\tilde{c} = 1$, $b^2 - a\tilde{c} = -1 < 0$: **elliptic**. Linear; the right hand side $-\rho(x, y)$ does not depend on $\phi$, so the equation is inhomogeneous.

4. Pendulum. Only one independent variable, $t$: an **ODE** of second order. **Nonlinear**, because the unknown appears inside $\sin\theta$. For small angles, $\sin\theta \approx \theta$ gives the linear harmonic oscillator. The type classification applies only to PDEs.

5. Burgers, variables $(x, t)$. The term $u\,u_x$ is a product of the unknown with its own derivative: **nonlinear**. The type is determined by the second derivatives alone, and these appear linearly with a constant coefficient: $\nu\,u_{xx} - u_t - u\,u_x = 0$ gives $a = \nu$, $b = 0$, $\tilde{c} = 0$, $b^2 - a\tilde{c} = 0$: **parabolic**.

6. Richards, variables $(z, t)$. To see the structure we carry out the derivatives. On the left, by the chain rule,

$$\frac{\partial\theta(\psi)}{\partial t} = \frac{\mathrm{d}\theta}{\mathrm{d}\psi}\,\frac{\partial\psi}{\partial t} = C(\psi)\,\frac{\partial\psi}{\partial t},$$

with the specific moisture capacity $C = \mathrm{d}\theta/\mathrm{d}\psi$. On the right, by the product rule and the chain rule $\partial K/\partial z = (\mathrm{d}K/\mathrm{d}\psi)\,\partial\psi/\partial z$,

$$\frac{\partial}{\partial z}\left[K\left(\frac{\partial\psi}{\partial z} + 1\right)\right] = K\frac{\partial^2\psi}{\partial z^2} + \frac{\mathrm{d}K}{\mathrm{d}\psi}\frac{\partial\psi}{\partial z}\left(\frac{\partial\psi}{\partial z} + 1\right).$$

The equation is

$$K(\psi)\,\psi_{zz} - C(\psi)\,\psi_t + \frac{\mathrm{d}K}{\mathrm{d}\psi}\left(\psi_z^2 + \psi_z\right) = 0.$$

It is **nonlinear** for three reasons: the coefficients $K$ and $C$ depend on the unknown $\psi$, and $\psi_z^2$ is a square of a derivative. Order 2. The only second derivative is $\psi_{zz}$ with $a = K(\psi) > 0$, $b = 0$, $\tilde{c} = 0$: $b^2 - a\tilde{c} = 0$, **parabolic**. The "+1" is gravity: with $z$ pointing upward, the water flux $-K\,\partial(\psi + z)/\partial z$ is directed downward even without a gradient of $\psi$.

Half a point for each row.